

differentiation, and translation properties of the Laplace transformation.

- Proper rational functions with simple poles can be expanded by partial fraction to obtain inverse Laplace transforms. Simple real poles lead to exponential waveforms and simple complex poles to damped sinusoids. Partial-fraction expansions of improper rational functions and functions with multiple poles require special treatment. MATLAB software can be used to find Laplace transforms from $f(t)$ and waveforms from $F(s)$.

- Using Laplace transforms to find the response of a circuit involves transforming the circuit differential equation into the s domain, algebraically solving for the response transform, and performing the inverse transformation to obtain the response waveform.
- The initial and final value properties determine the initial and final values of a waveform $f(t)$ from the value of $sF(s)$ as $s \rightarrow \infty$ and $s \rightarrow 0$, respectively. The initial value property applies if $F(s)$ is a proper rational function. The final value property applies if the poles of $sF(s)$ are in the left half plane.

PROBLEMS

OBJECTIVE 9-1 LAPLACE TRANSFORM (SECTS. 9-1, 9-2, 9-3)

Find the Laplace transform of a given signal waveform using transform properties and pairs, using the integral definition of the Laplace transformation, or using software applications. Locate the poles and zeros of the transform and construct a pole-zero diagram.

See Examples 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-7, 9-8, 9-9, 9-10 and Exercises 9-1, 9-2, 9-3, 9-4, 9-5, 9-6, 9-7, 9-8, 9-9, 9-10, 9-11, 9-12

- 9-1 Find the Laplace transform of $f(t) = 10[e^{-50t} - 2e^{-100t}]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-2 Find the Laplace transform of $f(t) = A[(3 - \alpha t)e^{-\alpha t}]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-3 Find the Laplace transform of $f(t) = 25[1 - 3 \cos(500t)]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-4 Find the Laplace transform of $f(t) = A[\cos(\beta t) - \sin(\beta t)]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-5 Find the Laplace transform of $f(t) = 28(t) - 100e^{-10t} \cos(t)u(10t)$. Locate the poles and zeros of $F(s)$.

- 9-6 Find the Laplace transform of $f(t) = 30[2 - 5t - 2e^{-5t}]u(t)$. Locate the poles and zeros of $F(s)$.

- 9-7 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:

- (a) $f_1(t) = [15e^{-5t} - 20e^{-10t}]u(t)$
(b) $f_2(t) = [5 \cos 10t + 10 \cos 20t]u(t)$

- 9-8 Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.

- (a) $f_1(t) = 48(t) + [20e^{-20t} + 40e^{-40t}]u(t)$
(b) $f_2(t) = [15 + 15 \cos 500t]u(t)$

- 9-9 Use MATLAB to find the Laplace transforms of the following waveforms. Locate the poles and zeros of $F(s)$.

- (a) $f_1(t) = \delta(t) + (625t e^{-25t})u(t)$
(b) $f_2(t) = [10 + e^{-10t} + 5 \cos 10t + 5 \sin 10t]u(t)$

- 9-10 Find the Laplace transforms of the following waveforms:

- (a) $f_1(t) = 38(t - 2)$
(b) $f_2(t) = 10e^{-50(t-1)}u(t-1)$
(c) $f_3(t) = 10e^{-50(t-2)}u(t-2)$

- 9-11 Use MATLAB to find the Laplace transform of the following waveform.

$$f(t) = [10 + 2e^{-10t}]u(t) + [5 \cos 100(t - 0.05)]u(t - 0.05)$$

- 9-12 Find the Laplace transform of the following waveform:

$$(a) f_1(t) = \frac{d}{dt}(100e^{-5000t} \cos 20,000t)u(t)$$

$$(b) f_2(t) = \int_0^t 50e^{-50x} dx + 100u(t) + 5 \frac{de^{-50t}}{dt} u(t)$$

- 9-13 (a) Write an expression for the waveform $f(t)$ in Fig. P9-13 using step and ramp functions.

- (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
(c) Verify the Laplace transform found in part (b) by applying the definition of the Laplace transformation, Eq. (9-2) to the waveform $f(t)$ found in part (a).

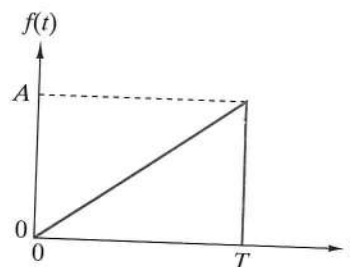


FIGURE P9-13

- (a) Write an expression for the waveform $f(t)$ in Fig. P9-14 using step functions.
 (b) Use the time-domain translation property to find the Laplace transform of the waveform $f(t)$ found in part (a).
 (c) Verify the Laplace transform found in part (b) by applying the definition of the Laplace transformation in Eq. (9-2) to the waveform $f(t)$ found in part (a).

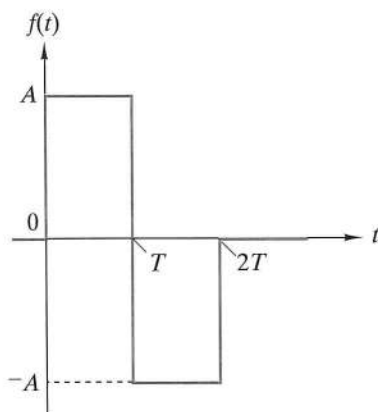


FIGURE P9-14

- (a) Find the Laplace transform of the following function. Locate the poles and zeros of $F(s)$.

$$f(t) = [500 + 75e^{-200t}t \sin 200t]u(t)$$

- (b) Validate your result using MATLAB.

EXERCISES 9-2 INVERSE TRANSFORMS PROBS. 9-4, 9-5

Find the inverse transform of a given Laplace transform using partial fraction expansion, using basic transform properties, or using software applications.

Examples 9-11, 9-12, 9-13, 9-14 and Exercises 9-13, 9-14, 9-16, 9-17, 9-18, 9-19, 9-20, 9-21

- Find the inverse Laplace transforms of the following functions:

$$F_1(s) = \frac{s+100}{s(s+50)}$$

$$F_2(s) = \frac{s+10}{s(s+50)(s+100)}$$

- Find the inverse Laplace transforms of the following functions:

$$F_1(s) = \frac{50(s+10)}{(s+5)(s+20)}$$

$$F_2(s) = \frac{2s^2}{(s+200)(s+500)}$$

- Find the inverse Laplace transforms of the following functions:

$$(a) F_1(s) = 400 \frac{1}{(s+10)^2 + 20^2}$$

$$(b) F_2(s) = \frac{2(s+10)}{(s+10)^2 + 20^2}$$

- 9-19 Find the inverse Laplace transforms of the following functions and sketch their waveforms for $\beta > 0$:

$$(a) F_1(s) = \frac{\beta(s+\beta)}{s(s^2+\beta^2)}$$

$$(b) F_2(s) = \frac{s(s+\beta)}{s^2+\beta^2}$$

- 9-20 Find the inverse Laplace transforms of the following functions:

$$(a) F_1(s) = \frac{\alpha^2}{s^2(s+\alpha)}$$

$$(b) F_2(s) = \frac{\alpha^2}{s(s+\alpha)^2}$$

- 9-21 Find the inverse Laplace transforms of the following functions:

$$(a) F_1(s) = \frac{600}{(s+10)(s+20)(s+40)}$$

$$(b) F_2(s) = \frac{6(s+10)}{(s+5)(s+20)}$$

- 9-22 Use MATLAB to find the inverse transforms of the following functions:

$$(a) F_1(s) = \frac{16s}{(s+2)(s^2+12s+13)}$$

$$(b) F_2(s) = \frac{4(s^2+4)}{s(s^2+16)}$$

- 9-23 Find the inverse transforms of the following functions:

$$(a) F_1(s) = \frac{(s+4000)(s+8000)}{s(s+2000)(s+6000)}$$

$$(b) F_2(s) = \frac{3s^4 + 10s^2 + 4}{s(s^2+1)(s^2+4)}$$

- 9-24 Find the inverse transforms of the following functions:

$$(a) F_1(s) = \frac{300(s+20)}{s(s^2+40s+300)}$$

$$(b) F_2(s) = \frac{200s}{(s+40)(s^2+4s+8)}$$

- 9-25 Find the inverse transforms of the following functions:

$$(a) F_1(s) = \frac{4(s^2+16)}{s(s^2+8s+32)}$$

$$(b) F_2(s) = \frac{2(s^2+30s+800)}{s(s^2+50s+400)}$$

- 9-26 Find the inverse transforms of the following functions and validate your answers using MATLAB:

$$(a) F_1(s) = \frac{(s+50)^2}{(s+20)^2(s+100)}$$

$$(b) F_2(s) = \frac{(s+20)^2}{(s+50)^2(s+100)}$$

9-27 A certain transform has a simple pole at $s = -20$, a simple zero at $s = -\gamma$, and a scale factor of $K = 1$. Select values for γ so the inverse transform is

- (a) $f(t) = \delta(t) - 5e^{-20t}$
- (b) $f(t) = \delta(t)$
- (c) $f(t) = \delta(t) + 5e^{-20t}$

9-28 Find the inverse transforms of the following functions:

- (a) $F_1(s) = \frac{e^{-3s}(s+20)}{(s+10)(s+30)}$
- (b) $F_2(s) = \frac{se^{-3s} + 20}{(s+10)(s+30)}$

 **9-29** Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{100s(s^2 + 30s + 400)}{(s+25)(s^3 + 6s^2 + 16s + 16)}$$

 **9-30** Use MATLAB to find the inverse transform and plot the poles and zeros of the following function:

$$F(s) = \frac{50(s^3 + 2s^2 + s + 2)}{s(s^3 + 4s^2 + 4s + 16)}$$

OBJECTIVE 9-3 CIRCUIT RESPONSE USING LAPLACE TRANSFORMS (SECT. 9-6)

Given a first- or second-order circuit:

- (a) Determine the circuit differential equation and the initial conditions (if not given).
- (b) Transform the differential equation into the s domain and solve for the response transform.
- (c) Use the inverse transformation to find the response waveform.

See Examples 9-15, 9-16, 9-17, 9-18 and Exercises 9-22, 9-23, 9-24, 9-25

9-31 Use the Laplace transformation to find the $v(t)$ that satisfies the following first-order differential equations:

- (a) $500 \frac{dv(t)}{dt} + 2500v(t) = 0, \quad v(0-) = 100 \text{ V}$
- (b) $\frac{dv(t)}{dt} + 200v(t) = 400u(t) \text{ V}, \quad v(0-) = -100 \text{ V}$

9-32 Use the Laplace transformation to find the $i(t)$ that satisfies the following first-order differential equation:

$$\frac{di(t)}{dt} + 500i(t) = [0.100e^{-200t}]u(t) \text{ A}, \quad i(0-) = 0 \text{ A}$$

9-33 The switch in Figure P9-33 has been open for a long time and is closed at $t = 0$. The circuit parameters are $R = 400 \Omega$, $L = 330 \text{ mH}$, and $V_A = 15 \text{ V}$.

- (a) Find the differential equation for the inductor current $i_L(t)$ and the initial condition $i_L(0)$.
- (b) Solve for $i_L(t)$ using the Laplace transformation.

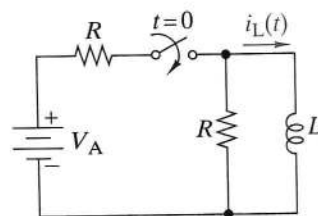


FIGURE P9-33

9-34 The switch in Figure P9-33 has been opened for a long time and is closed at $t = 0$. The circuit parameters are $R = 200 \Omega$, $L = 200 \text{ mH}$, and $V_A = 50e^{-1000t} \text{ V}$.

- (a) Find the differential equation for the inductor current $i_L(t)$ and initial condition $i_L(0)$.
- (b) Solve for $i_L(t)$ using the Laplace transformation.

9-35 The switch in Figure P9-35 has been open for a long time. At $t = 0$ the switch is closed.

- (a) Find the differential equation for the capacitor voltage $v_C(t)$ and initial condition $v_C(0)$.
- (b) Find $v_C(t)$ using the Laplace transformation for $25u(t) \text{ V}$.

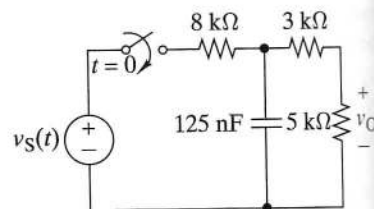


FIGURE P9-35

9-36 Repeat Problem 9-35 for the input $v_S(t) = 100[\cos 1000t]u(t) \text{ V}$.

9-37 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2v(t)}{dt^2} + 25 \frac{dv(t)}{dt} + 100v(t) = 0, \quad v(0-) = 10 \text{ V and } \frac{dv(0-)}{dt} = 0$$

9-38 Use the Laplace transformation to find the $v(t)$ that satisfies the following second-order differential equation:

$$\frac{d^2v(t)}{dt^2} + 100 \frac{dv(t)}{dt} + 500v(t) = 0, \quad v(0-) = 0 \text{ and } \frac{dv(0-)}{dt} = 100 \text{ V/s}$$

9-39 The switch in Figure P9-39 has been open for a long time and is closed at $t = 0$. The circuit parameters are 500Ω , $L = 2.5 \text{ H}$, $C = 2.5 \mu\text{F}$, and $V_A = 20 \text{ V}$.

- (a) Find the circuit differential equation in $i_L(t)$ and initial conditions $i_L(0)$ and $v_C(0)$.
- (b) Use Laplace transforms to solve for $i_L(t)$ for $t \geq 0$.

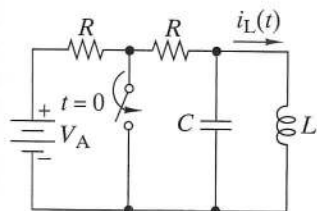


FIGURE P9-39

Repeat Problem 9-39 with $R = 5 \text{ k}\Omega$.

The switch in Figure P9-41 has been closed for a long time and is opened at $t = 0$.

Find the circuit differential equation in $v_C(t)$ and the initial conditions $i_L(0)$ and $v_C(0)$.

The circuit parameters are $L = 5 \text{ H}$, $C = 2.5 \mu\text{F}$, $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, and $v_S = 20u(t) \text{ V}$. Use Laplace transforms and MATLAB to solve for $v_C(t)$ for $t \geq 0$.

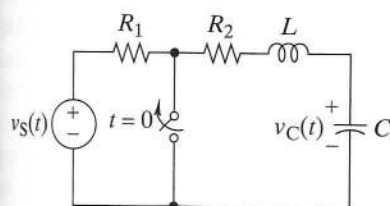


FIGURE P9-41

Repeat Problem 9-41 with $R_2 = 500 \Omega$.

The RLC circuit in Figure P9-43 is in the zero state when $t = 0$ an exponential source, $v_S(t) = V_A e^{-at} \text{ V}$, is suddenly connected to it.

Find the circuit integrodifferential equation that describes the behavior of the current in the circuit.

If $R = 100 \Omega$, $L = 100 \text{ mH}$, $C = 10 \mu\text{F}$, $V_A = 10 \text{ V}$, and $a = 500 \text{ s}^{-1}$, use Laplace transforms and MATLAB to solve for $i(t)$ for $t \geq 0$.

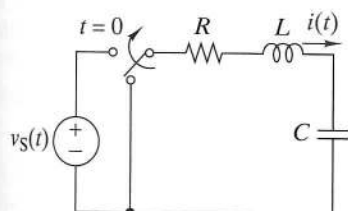


FIGURE P9-43

Repeat Problem 9-43 for an exponential source, $v_S(t) = V_A e^{-500t} \text{ V}$, that is suddenly connected to the circuit.

PROBLEM 9-44 INITIAL AND FINAL VALUE PROPERTIES (SECT. 9-7)

Use the initial and final value properties of Laplace transforms to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

Examples 9-19, 9-20 and Exercise 9-26

9-45 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

$$(a) F_1(s) = \frac{16s}{(s+2)(s^2+12s+13)}$$

$$(b) F_2(s) = \frac{s+10}{s(s+50)(s+100)}$$

9-46 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the transforms below. If either property is not applicable, explain why.

$$(a) F_1(s) = 400 \frac{1}{(s+10)^2 + 20^2}$$

$$(b) F_2(s) = \frac{3s^4 + 10s^2 + 4}{s(s^2+1)(s^2+4)}$$

9-47 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the following transforms. If either property is not applicable, explain why.

$$(a) F_1(s) = \frac{50(s^2+5s+6)}{(s+2)(s+6)(s+12)}$$

$$(b) F_2(s) = \frac{10(s^2+10s+40)}{s(s^2-625)}$$

9-48 Use the initial and final value properties to find the initial and final values of the waveforms corresponding to the following transforms. If either property is not applicable, explain why.

$$(a) F_1(s) = \frac{12s(s+6)}{s^2+6s+9}$$

$$(b) F_2(s) = \frac{20(s^2+10s-20)}{s(s^2+100)}$$

9-49 Use the initial and final value properties to find the initial and final values of the waveform corresponding to the following transform. If either property is not applicable, explain why.

$$F(s) = \frac{80(s^3+2s^2+s+2)}{s(s^3+4s^2+4s+16)}$$

9-50 The MATLAB function `limit` can be used to take the limit of a symbolic expression. Use MATLAB and the initial and final value properties to find the initial and final values of the waveforms corresponding to the following transforms. If either property is not applicable, explain why. Use MATLAB again to compute the waveforms corresponding to each transform, and then take limits in the time domain to verify the answers found using the initial and final value properties.

$$(a) F_1(s) = \frac{(s+50)^2}{(s+20)^2(s+100)}$$

$$(b) F_2(s) = \frac{(s+20)^2}{(s+50)^2(s+100)}$$

SUMMARY

Kirchhoff's laws apply to voltage and current waveforms in the time domain and to the corresponding transforms in the s domain.

The s -domain models for the passive elements include initial condition sources and the element impedance or admittance. Impedance is the proportionality factor in the expression $V(s) = Z(s)I(s)$ relating the voltage and current transforms. Admittance is the reciprocal of impedance.

The impedances of the three passive elements are $Z_R(s) = R$, $Z_L(s) = Ls$, and $Z_C(s) = 1/Cs$.

The s -domain circuit analysis techniques closely parallel the analysis methods developed for resistance circuits. Basic analysis techniques, such as circuit reduction, Thévenin's and Norton's theorems, the output method, or superposition, can be used in simple circuits. More complicated networks require a general approach, such as the node-voltage or mesh-current methods.

Response transforms are rational functions whose poles are zeros of the circuit determinant or poles of the transform of the input driving forces. Poles

introduced by the circuit determinant are called natural poles and lead to the natural response. Poles introduced by the input are called forced poles and lead to the forced response.

- In linear circuits, response transforms and waveforms can be separated into zero-state and zero-input components. The zero-state component is found by setting the initial capacitor voltages and inductor currents to zero. The zero-input component is found by setting all input driving forces to zero.
- The main purpose of s -domain circuit analysis is to gain insight into circuit performance without necessarily finding the time-domain response. The natural poles reveal the form, stability, and observability of the circuit's response. The number of natural poles is never greater than the number of energy storage elements in the circuit.
- A circuit is stable if all of its natural poles are in the left half of the s plane. Passive circuits are inherently stable so their natural poles are in the left half plane. Active circuits can be stable when circuit parameters are in one range and unstable for parameters outside this range.

PROBLEMS

OBJECTIVE 10-1 EQUIVALENT IMPEDANCE (EXERCISES 10-1, 10-2)

In a linear circuit, use series and parallel equivalence to find the equivalent impedance at specified terminal pairs. Use element values to obtain specified pole locations.

Examples 10-2, 10-4, 10-5, 10-6 and Exercises 10-3, 10-5

Find $Z_{EQ}(s)$ in Figure P10-1. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

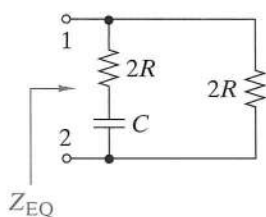


FIGURE P10-1

Find $Z_{EQ}(s)$ in Figure P10-2. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

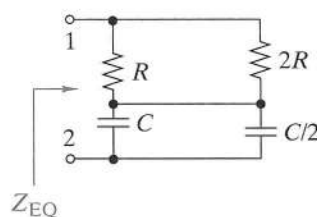


FIGURE P10-2

10-3 Find $Z_{EQ}(s)$ in Figure P10-3. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros for $R = 2\text{ k}\Omega$, $L = 0.4\text{ H}$, and $C = 100\text{ nF}$.

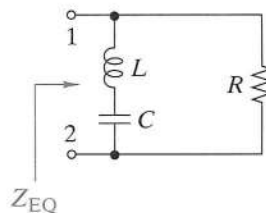


FIGURE P10-3

- 10-4 Find $Z_{EQ}(s)$ in Figure P10-4. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

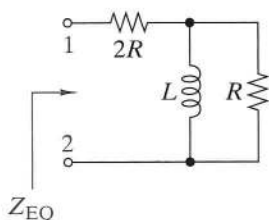


FIGURE P10-4

- 10-5 Find $Z_{EQ}(s)$ in Figure P10-5. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

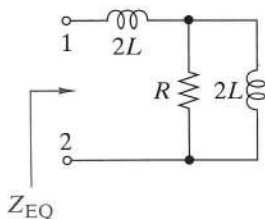


FIGURE P10-5

- 10-6 Find the poles and zeros of $Z_{EQ}(s)$ in Figure P10-6 with $R = 1 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 500 \text{ nF}$.

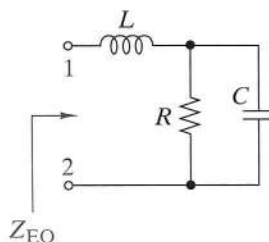


FIGURE P10-6

- 10-7 Find $Z_{EQ1}(s)$ and $Z_{EQ2}(s)$ in Figure P10-7. Express each impedance as a rational function and locate its poles and zeros.

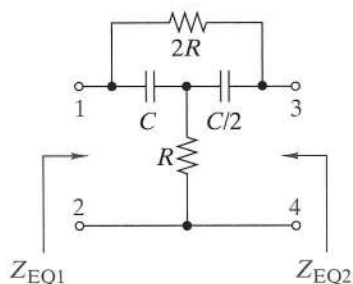


FIGURE P10-7

- 10-8 Find $Z_{EQ}(s)$ in Figure P10-8. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

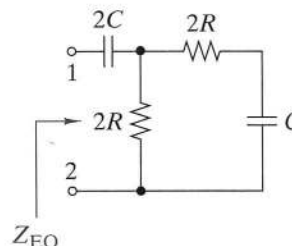


FIGURE P10-8

- 10-9 Find $Z_{EQ1}(s)$ and $Z_{EQ2}(s)$ in Figure P10-9. Express each impedance as a rational function and locate its poles and zeros.

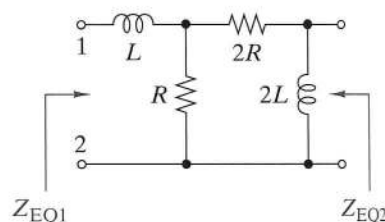


FIGURE P10-9

- 10-10 Find the equivalent impedance between terminals 1 and 2 in Figure P10-10. Select values of R and L so that $Z_{EQ}(s)$ has a pole at $s = -2000 \text{ rad/s}$. Locate the zeros of $Z_{EQ}(s)$ for your choice of R and L .

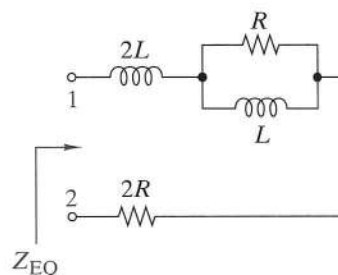


FIGURE P10-10

OBJECTIVE 10-2 BASIC CIRCUIT ANALYSIS TECHNIQUES (SECTS. 10-2, 10-3)

Given a linear circuit,

- Determine the initial conditions (if not given) and transform the circuit into the s domain.
- Solve for zero-state and/or zero-input response transients and waveforms using basic analysis methods such as circuit reduction, unit output, Thévenin/Norton equivalent circuits, or superposition.

(c) Locate the poles and zeros of $Z_{EQ}(s)$ to place the circuit into the s domain. See Examples 10-9 and Exercise 10-8, 10-9, 10-10.

10-11 The switch in the circuit is closed at $t = 0$. Find the long-time and short-time values of the current $i_L(t)$ in symbolic form.

10-12 The switch in the circuit is closed at $t = 0$. Find the long-time and short-time values of the current $i_L(t)$ in symbolic form.

10-13 The switch in the circuit is closed at $t = 0$. Find the long-time and short-time values of the current $i_L(t)$ in symbolic form.

10-14 The switch in the circuit is closed at $t = 0$. Find the long-time and short-time values of the current $i_L(t)$ in symbolic form.

10-15 The switch in the circuit is closed at $t = 0$. Find the long-time and short-time values of the current $i_L(t)$ in symbolic form.